FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Paediatric Emergencies, Triage & ETAT

Lecture | Week 3 · Fits, TB, kidneys & nutrition | Paediatrics | Dr Aliku

Most children who die in Ugandan hospitals die within 24 hours of arrival, often while waiting in the queue; ETAT (Emergency Triage, Assessment and Treatment) sorts every sick child within minutes into Emergency (ABCD signs, treat now), Priority (front of the queue) or Queue, and treats airway, breathing, circulation, coma, convulsions and dehydration by weight-based protocols before any diagnosis is made.

Must know before the ward round

1. Triage every child on arrival, before registration or payment, in under 1 minute per child: E = Emergency (treat now), P = Priority (see next), Q = Queue (non-urgent).
2. Emergency signs = ABCD: obstructed/absent breathing, severe respiratory distress, central cyanosis; shock (cold hands + capillary refill >3 s + weak fast pulse); coma (AVPU = P or U) or convulsing; severe dehydration (any 2 of lethargy, sunken eyes, very slow skin pinch) in a child with diarrhoea.
3. Priority signs = 3 TPR-MOB: Tiny baby (<2 months), Temperature very high, Trauma; Pallor (severe), Poisoning, Pain (severe); Respiratory distress, Restless/irritable/lethargic, Referral (urgent); Malnutrition (visible severe wasting), Oedema of both feet, Burns (major).
4. Oxygen: give to any child with emergency signs whose SpO_2 is <94%; once stable, give if SpO_2 <90%. Nasal prongs 0.5–1 L/min (neonate), 1–2 L/min (infant), 2–4 L/min (older child).
5. Shock (not SAM): 10–20 mL/kg Ringer's lactate or 0.9% saline over 30–60 min (WHO ETAT 2016), reassess after each bolus, stop at signs of overload; transfuse 20 mL/kg whole blood if severe anaemia.
6. Shock in SAM: 15 mL/kg over 1 hour of Ringer's lactate with 5% dextrose (or half-strength Darrow's with 5% glucose); repeat once if improving, then switch to ReSoMal; if no improvement after two boluses, maintenance fluid + transfuse 10 mL/kg over 3 h.
7. Hypoglycaemia (<2.5 mmol/L, or <3.0 mmol/L in SAM): 10% dextrose 5 mL/kg IV (neonate 2 mL/kg), recheck in 30 min. If you cannot measure glucose, treat anyway in every child with coma or convulsions.
8. Convulsing: diazepam 0.5 mg/kg rectally or 0.2–0.3 mg/kg slow IV (max 10 mg), or buccal midazolam 0.3–0.5 mg/kg; repeat once after 10 min; if still fitting, phenobarbital 15–20 mg/kg IV/IM over 15–20 min (neonates: phenobarbital 20 mg/kg first).
9. Severe dehydration (Plan C): Ringer's lactate 100 mL/kg; infant <12 months 30 mL/kg in 1 h then 70 mL/kg in 5 h; child ≥12 months 30 mL/kg in 30 min then 70 mL/kg in 2½ h. Not in SAM.
10. Anaphylaxis: adrenaline 1 mg/mL (1:1000) 0.01 mL/kg IM in the anterolateral thigh (max 0.5 mL), repeat every 5 min if needed; then 10 mL/kg crystalloid bolus if shocked.
11. Paediatric BLS: 5 rescue breaths, then 15 compressions : 2 breaths at 100–120/min, depth one-third of the chest (about 4 cm infant, 5 cm child); call for help after 1 minute if alone.
12. Weight if no scale: 1–12 months = (0.5 × age in months) + 4; 1–5 years = (2 × age) + 8; 6–12 years = (3 × age) + 7 kg (APLS). Weigh as soon as the child is stable.

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1. Definitions and Key Concepts

ETAT (Emergency Triage, Assessment and Treatment) is the WHO system for making sure that the sickest child in the waiting area is seen **first**, not last. It has three parts:

1. **Triage**: a quick look at every child on arrival (no history, no stethoscope needed) to sort into **E**mergency, **P**riority or **Q**ueue.
2. **Assessment**: for emergency cases, the **ABCD** approach (Airway and Breathing; Circulation; Coma and Convulsions; Dehydration).
3. **Treatment**: start protocol treatment immediately, **before a diagnosis** is made. Then take the history and examine fully.

> **CLINICAL PEARL**
> The one thing never to forget: **treat what is killing the child, in ABCD order, before you work out why.** A convulsing child gets glucose and diazepam before you ask about the fever; a child in shock gets fluids before you send the malaria test. And **always check the glucose** in a child who is unconscious or convulsing.

Term	Meaning
Triage	Sorting patients by **urgency** of need, not by order of arrival. ETAT triage takes **15–20 seconds per child** and should be done by any trained health worker (nurse, clinician, even trained porters or records staff)
Emergency (E)	A sign that the child may die within minutes or hours unless treated: **ABCD emergency signs**. Treat immediately in the resuscitation area
Priority (P)	Signs that the child is at high risk and must be **assessed next**, ahead of the queue, but does not need resuscitation now
Queue (Q)	Non-urgent; wait their turn. Re-triage if the child deteriorates in the queue
ABCD	Airway & Breathing; Circulation; Coma & Convulsions; Dehydration (severe)
Shock (WHO)	**All three**: cold hands **and** capillary refill **>3 seconds and** weak, fast pulse
Impaired circulation	One or two of the shock signs, not all three: do **not** give a bolus, but give maintenance fluids and watch closely
AVPU	**A**lert, responds to **V**oice, responds to **P**ain, **U**nresponsive. **P or U = coma** (roughly GCS ≤8)
Severe dehydration (in diarrhoea)	Two or more of: **lethargy/unconscious**, **sunken eyes**, **very slow skin pinch (≥2 s)**, unable to drink or drinks poorly
Danger signs (IMCI)	Used in the outpatient/IMCI setting: **unable to drink or breastfeed, vomits everything, convulsions (now or in this illness), lethargic or unconscious**. Any one = refer urgently / treat as severe

Emergency signs (ABCD)

Letter	Emergency sign	How you check it in seconds
Airway & **B**reathing	Obstructed or absent breathing; **severe respiratory distress**; **central cyanosis**	Look, listen, feel; grunting, severe chest indrawing, head nodding, inability to feed due to breathing; blue tongue/lips

Letter	Emergency sign	How you check it in seconds
Circulation	**Signs of shock**: cold hands + CRT >3 s + weak fast pulse	Touch the hand; press the nail or sternum 3 s; feel the radial (brachial in infants) pulse
Coma / **C**onvulsions	**Coma** (AVPU P or U) or **convulsing now**	Call, then pinch; look for jerking, eye deviation, rhythmic movements
Dehydration	**Severe dehydration** in a child with **diarrhoea**: 2 of lethargy, sunken eyes, very slow skin pinch	Only assess D if the child has diarrhoea

Priority signs: "3 TPR-MOB"

Group	Sign
Ts	**T**iny baby (any sick infant <2 months); **T**emperature very high (≥39.5 °C) or very low; **T**rauma or other urgent surgical condition
Ps	**P**allor (severe); **P**oisoning (or history of ingestion); **P**ain (severe)
Rs	**R**espiratory distress (not severe); **R**estless, continuously irritable or lethargic; **R**eferral (urgent, from another facility)
M	**M**alnutrition: visible severe wasting
O	**O**edema of both feet
B	**B**urns (major)

2. Burden and Uganda Context

- Studies from Malawi and Kenya in the 1990s and 2000s showed that many hospital deaths happened in the **first 24 hours**, often before a doctor had seen the child. Introducing ETAT at Lilongwe Central Hospital roughly **halved inpatient mortality**. Uganda adopted ETAT through the Ministry of Health, and it is taught at Mulago, Mengo and regional referral hospitals (RRHs).
- Uganda's under-5 mortality was about **52 per 1,000 live births** in UDHS 2022 (down from 64 in 2016). The leading killers of children admitted to Ugandan hospitals are **severe malaria (with severe anaemia), pneumonia, diarrhoea with dehydration, neonatal conditions, severe acute malnutrition and sepsis**, often in combination and often with HIV.
- Many paediatric deaths happen within **24–48 hours of admission**. Delays happen at home (care-seeking), on the road (transport) and **inside the hospital** (registration, payment, waiting for the clinician, waiting for drugs). **Triage attacks the third delay.**
- Children arrive late and very sick. A typical Mengo or Mulago acute care unit sees children with **cerebral malaria convulsing, severe anaemia with Hb 3 g/dL, and severe pneumonia** within the same hour.

> **UGANDA CONTEXT**
> Realities of ETAT in Uganda: there may be **one nurse** triaging a full waiting area; **pulse oximeters** and **oxygen concentrators** have become common at RRHs and many HC IVs since COVID-19, but power cuts limit concentrators; **glucometers and strips** run out; **blood** comes from the Uganda Blood Transfusion Service and is often short; **rectal diazepam** is usually available, while **buccal midazolam** is not always stocked. At HC III level you stabilise (oxygen, glucose, first-dose antibiotics and antimalarial, diazepam, fluids) and **refer with a note and a health worker**. Mulago's Acute Care Unit and Mengo's paediatric emergency area run ETAT-based triage at the door.

3. Principles and Framework

The ETAT pathway

1. **Arrival → triage** (15–20 s), before registration or payment.
2. **Any emergency sign?** Yes → resuscitation area, **ABCD now**:
   - **A**: open the airway / clear obstruction.
   - **B**: oxygen / bag-mask.
   - **C**: shock? IV/IO, fluids per protocol (normal vs SAM).
   - **C**: coma/convulsions: glucose, anticonvulsant, recovery position.
   - **D**: severe dehydration: Plan C.
   - Then: history, full examination, investigations, specific treatment, admit.
3. No emergency sign → **any priority sign?** Yes → front of the queue, assess next.
4. Neither → **queue**; re-triage if the child deteriorates.

At every letter: **assess, treat, reassess** before moving on.

Why children decompensate suddenly

- **Airway**: the airway is small; 1 mm of oedema in an infant's subglottis cuts the cross-sectional area by more than half. The tongue is large, the occiput prominent (so an infant's neck flexes when lying flat), and infants are preferential nose breathers.
- **Breathing**: children have a high metabolic rate and oxygen use per kg, small functional residual capacity and fatigable respiratory muscles. A child with severe respiratory distress can **tire and stop breathing** quite suddenly. **Bradycardia in a hypoxic child is pre-arrest.**
- **Circulation**: cardiac output depends mainly on **heart rate**. Children maintain blood pressure by vasoconstriction until they have lost about 30–40% of circulating volume, so **hypotension is a late, pre-terminal sign**. Tachycardia, cold peripheries and prolonged capillary refill come first.
- **Glucose**: small glycogen stores and high glucose use per kg mean that fasting, fever, malaria, sepsis and quinine quickly cause **hypoglycaemia**, which itself causes coma and convulsions.
- **Most paediatric cardiac arrests are secondary** to respiratory failure or shock (not primary cardiac rhythms as in adults). That is why paediatric BLS starts with **5 rescue breaths**, and why recognising and treating A, B and C early prevents arrest.

The FEAST lesson: why fluids are given carefully

The **FEAST trial** (Maitland et al., NEJM 2011), done largely in **Uganda** (including Mbale and Soroti), Kenya and Tanzania, randomised febrile children with impaired perfusion (mostly malaria and sepsis, not diarrhoea) to fast fluid boluses (20–40 mL/kg) or no bolus. **Bolus fluids increased 48-hour mortality** (about 10.6% vs 7.3%), mainly from cardiovascular collapse. The result changed practice:

- **No boluses** for children with impaired circulation who do **not** meet the full WHO shock definition.
- For true shock, **10–20 mL/kg over 30–60 min**, with reassessment (WHO ETAT 2016).
- Severe dehydration from diarrhoea is different: these children have lost fluid and need it back (Plan C).
- SAM and severe anaemia are special cases (section 6).

Age-specific vital signs

Age	Fast breathing (WHO)	Heart rate (normal, awake)	Minimum systolic BP
<2 months	**≥60/min**	110–160	60 mmHg (term neonate)

Age	Fast breathing (WHO)	Heart rate (normal, awake)	Minimum systolic BP
2–12 months	**≥50/min**	110–160	70 mmHg
1–5 years	**≥40/min**	95–140	70 + (2 × age in years)
5–12 years	≥30/min	80–120	70 + (2 × age) up to 90
>12 years	≥20/min	60–100	90 mmHg

Weight when there is no scale (APLS)

Age	Estimated weight
1–12 months	(0.5 × age in months) + 4 kg
1–5 years	**(2 × age) + 8 kg**
6–12 years	(3 × age) + 7 kg

Use a length-based tape (Broselow) if available, and **weigh as soon as the child is stable**.

4. Application in Practice

Step 0: First 5 minutes

1. **Triage at the door**: any emergency sign → carry the child straight to the resuscitation area. Shout for help (nurse, senior clinician).
2. **A**: open the airway (head tilt–chin lift in a child; neutral position in an infant, towel under the shoulders). If trauma is possible, jaw thrust and stabilise the neck. If choking, back blows and chest/abdominal thrusts (section 6).
3. **B**: if not breathing or gasping → **bag-mask ventilation** (5 rescue breaths). If breathing with distress or cyanosis → **oxygen**; target SpO_2 ≥94% while emergency signs are present.
4. **C**: check the three shock signs. Stop any external bleeding. Insert an IV or **intraosseous (IO)** line (proximal tibia, if IV fails within 90 seconds or 2 attempts). Draw blood for **glucose, Hb, malaria test, grouping and cross-match**. Give fluids per protocol (normal vs SAM).
5. **C**: coma or convulsing → recovery position (if no trauma), **check glucose** and give dextrose if low (or if you cannot check), **anticonvulsant** if fitting.
6. **D**: diarrhoea with severe dehydration → start **Plan C** (or SAM protocol).
7. Keep warm, keep the mother close, **weigh** when possible.

> **RED FLAG**
> Call the most senior person available immediately for: apnoea or bradycardia, SpO_2 <90% despite oxygen, shock not responding to the first bolus, status epilepticus, coma with signs of raised ICP, or a sick neonate.

Step 1: Focused history

Taken **after** emergency treatment has started, usually from the mother or caregiver.

Question	Why it matters
Age; weight (or last weight on the child health card)	All doses and fluids are per kg; <2 months = tiny baby (priority, different causes and drugs)
Main complaint, onset and duration: fever, cough, fast breathing, diarrhoea/vomiting, convulsions, reduced consciousness, pallor, swelling	Acute (hours: poisoning, convulsions) vs subacute (days: malaria, pneumonia, meningitis); points to malaria, pneumonia, gastroenteritis, meningitis, severe anaemia, SAM

Question	Why it matters
Convulsions: how many, how long, focal, first ever? Fever?	Febrile convulsion vs cerebral malaria vs meningitis vs epilepsy vs hypoglycaemia
Fluid in and out: drinking or breastfeeding? Vomiting everything? Diarrhoea frequency, blood? Last urine?	Dehydration severity and plan; danger signs
Cough, fast or difficult breathing, noisy breathing	Pneumonia, asthma, foreign body, croup
Possible ingestion: paraffin, drugs, pesticides, herbs, local brew	Poisoning changes treatment; avoid harmful decontamination (Common Poisoning topic)
Drugs given at home or at the clinic: antimalarials, quinine, herbal medicines, diazepam already given	Quinine causes hypoglycaemia; repeated benzodiazepines cause respiratory depression; count total doses
Immunisation (child health card)	Unimmunised child: measles, pertussis, tetanus, diphtheria, invasive Hib/pneumococcal disease
HIV status of child and mother; on cotrimoxazole/ART?	PCP, TB, severe bacterial infection; shapes antibiotics
Feeding and growth (SAM history, recent weight loss, oedema)	SAM changes fluid and feeding protocols
Known illness: sickle cell, heart disease, asthma, epilepsy, diabetes	Sickle crisis, heart failure (careful fluids), asthma, DKA

Step 2: Focused examination

The ETAT examination is ordered **A → B → C → D → (Disability) → Exposure**. Once the child is stable, complete the full examination in the order of Hutchison's Clinical Methods (24th edition), adapted for children.

ABCDE (during resuscitation)

1. **Airway**: is the child talking or crying? Stridor, gurgling, foreign body?
2. **Breathing**: respiratory rate (count a full minute), chest indrawing, grunting, nasal flaring, head nodding, wheeze, crackles, **SpO_2**, central cyanosis.
3. **Circulation**: hands warm or cold, capillary refill time, pulse rate and volume, **pallor** (palms, conjunctivae), heart rate, BP if a paediatric cuff exists, liver size (enlarged in heart failure and fluid overload).
4. **Coma/Convulsions**: AVPU (or Blantyre coma scale in under-5s), pupils, posture (decorticate/decerebrate), neck stiffness, bulging fontanelle, **glucose**.
5. **Dehydration**: eyes, skin pinch, drinking, lethargy.
6. **Exposure**: temperature, rashes (petechiae/purpura = meningococcaemia), **bilateral pitting oedema of the feet** and **severe wasting** (SAM), burns, injuries, jaundice.
7. **Weight** (and MUAC if 6–59 months) as soon as stable.

Full examination once stable (Hutchison order)

- **Introduction and consent**: introduce yourself, confirm the child's identity, explain to the mother and gain consent; **hand hygiene**; examine on the mother's lap where possible, exposing in stages and keeping the child warm.
- **General inspection**: well or unwell, AVPU, colour (pallor, cyanosis, jaundice), hydration, wasting, oedema, dysmorphism; **vitals** against the age table; **weight, MUAC and growth** plotted.
- **Hands, face and neck**: palmar pallor, CRT, clubbing; conjunctival pallor, jaundice, sunken eyes, nasal flaring; neck stiffness, lymph nodes, fontanelle.
- **Systems**: respiratory (rate, indrawing, grunting, stridor, wheeze, crackles, air entry, SpO_2); cardiovascular (pulse, CRT, hand temperature, heart sounds and gallop, liver size, BP); neurological (AVPU/Blantyre score, pupils, tone, posturing, focal signs); abdomen (distension, tenderness, hepatosplenomegaly: malaria, sickle cell, heart failure); skin (petechiae, purpura, rash, burns, skin pinch).

- **Distressing parts last** (ears, throat), then thank the mother, **hand hygiene**, summarise.

Document these positives	And these negatives
Time of arrival and triage category	"No stridor; no central cyanosis"
RR, SpO_2 (in air and on oxygen), HR, CRT, temperature	"Not in shock (warm hands, CRT <3 s)"
AVPU/Blantyre score, pupils, glucose value	"No neck stiffness, fontanelle not bulging"
Weight, MUAC, oedema	"No bilateral pedal oedema; not severely wasted"
Every drug and fluid with dose and time	"No signs of fluid overload after bolus"

Step 3: Bedside tests

- **Blood glucose** (glucometer): every child with coma, convulsions, SAM, shock, or who cannot feed.
- **Haemoglobin** (HemoCue) or packed cell volume: pallor, malaria, sickle cell.
- **Malaria rapid diagnostic test (mRDT)** or blood smear: every febrile child in Uganda.
- **Pulse oximetry**: every child with respiratory signs, shock or coma.
- **Bedside clotting test**: snakebite, suspected DIC.
- **Urine dipstick**: glucose and ketones (DKA), blood (haemoglobinuria in blackwater fever).
- **MUAC** (6–59 months): <11.5 cm = SAM.

Step 4: Laboratory and imaging

Test	When	Expected / abnormal finding	How it changes management	Availability
Blood glucose	Coma, convulsions, SAM, shock	<2.5 mmol/L (<3.0 in SAM); very high → DKA	10% dextrose 5 mL/kg	Glucometer at HC III upward (strips often out)
Hb / full blood count	Pallor, malaria, sepsis	Hb <4 g/dL, or <6 g/dL with distress	Transfuse; fluid strategy	HemoCue at HC IV; FBC at hospital
Malaria smear / mRDT	Any fever	Positive; parasitaemia	IV artesunate for severe malaria	Everywhere
Blood grouping & cross-match	Severe anaemia, bleeding	Group and compatible unit	Blood ordered early	Hospitals with blood banks
Blood culture, FBC, CRP	Sepsis, shock, meningitis	Organism and sensitivity	Narrow antibiotics later	Regional/national referral (often unavailable)
Lumbar puncture	Suspected meningitis once stable	Cloudy CSF, high WBC, low glucose	Antibiotic choice/duration	Hospitals; **delay if** raised ICP signs, shock or unstable
Electrolytes, urea, creatinine	Severe dehydration, shock, AKI, DKA	Na, K, urea abnormal	Fluid type, potassium	RRH/national referral
Blood gas	Shock, DKA, respiratory failure	Acidosis, high CO_2	Severity, ventilation	Mulago/private hospitals mainly
HIV test (child and mother)	Every sick child with unknown status	Positive	ART, cotrimoxazole, broader differential	Everywhere
Chest X-ray	Respiratory distress after stabilisation	Pneumonia, effusion, pneumothorax, foreign body, cardiomegaly	Drainage, specific treatment	Hospitals

Step 5: Putting it together

Model one-line summary: "A 2-year-old boy, 10 kg, triaged **Emergency** for **convulsions and coma**; glucose 1.8 mmol/L, mRDT positive, Hb 5.2 g/dL; seizures stopped after rectal diazepam and IV dextrose; **working diagnosis: severe malaria (cerebral malaria with hypoglycaemia and severe anaemia)**."

Problem list: (1) coma/convulsions, (2) hypoglycaemia, (3) severe malaria, (4) severe anaemia, (5) possible meningitis (LP once stable).

Model case presentation

> "Baby J.N., an **18-month-old boy** from Nansana, **weight 9.0 kg** (weight-for-height Z-score –1.5, MUAC 12.8 cm, no oedema), brought in at 10:40 by his mother with **3 days of fever and 1 day of reduced feeding, followed by a generalised convulsion this morning** lasting about 20 minutes at home. He was **triaged Emergency (convulsing and AVPU P)**. Glucose was **2.0 mmol/L**, mRDT positive, Hb 6.8 g/dL. He received **rectal diazepam 4.5 mg**, **10% dextrose 45 mL IV**, oxygen for SpO_2 92% and **IV artesunate 27 mg**. He has not convulsed again; now AVPU V, glucose 5.4 mmol/L. **Immunisation up to date for age** per the child health card (measles-rubella at 9 months given); **breastfeeding plus family foods**; mother **HIV-negative** in this pregnancy, child not tested. **Assessment: severe malaria (cerebral malaria with hypoglycaemia), moderate anaemia**, differential meningitis. **Plan**: continue artesunate, maintenance fluids with dextrose, 4-hourly glucose, Blantyre score, LP when stable, cross-match if Hb falls, HIV test."

NOTE

Worked example: is this shock? A 3-year-old (14 kg) with fever for 2 days: hands cool, CRT 4 s, radial pulse 170/min and weak, AVPU = V, Hb 9 g/dL, no diarrhoea. **Interpretation**: all three WHO signs are present → **shock**; not SAM, not severely anaemic. **Action**: oxygen, IV access, glucose, **Ringer's lactate 20 mL/kg (280 mL) over 30–60 min** (10 mL/kg if you are worried about the heart), reassess after the bolus; if still shocked give a second 10–20 mL/kg; start **ceftriaxone** and **artesunate** if mRDT positive. If the liver enlarges or crackles appear, stop the fluid.

5. Clinical Acumen

Thinking like a clinician

- **"Is this child in shock, or just cold and febrile?"** All three WHO signs are needed (cold hands, CRT >3 s, weak fast pulse). A febrile child with malaria may have one or two; FEAST shows boluses can kill these children. Give maintenance fluids and recheck in 15–30 minutes.
- **Pallor + respiratory distress + big liver in a child with malaria** is usually **severe anaemia with heart failure**, not pneumonia and not "shock needing saline". The treatment is **blood**, slowly, sometimes with furosemide, not crystalloid boluses.
- **Unconscious child: think "G before D"** (glucose before diazepam/diagnosis). Hypoglycaemia is common in cerebral malaria, sepsis, SAM and after quinine.
- **A child "fitting" with fever**: cerebral malaria, meningitis and febrile convulsion look alike at the door. Treat the seizure and glucose first; then decide on LP, mRDT/blood smear and antibiotics.
- **Deep sighing breathing with a clear chest** = metabolic acidosis (DKA, severe malaria, dehydration), not pneumonia.
- **SAM changes everything**: shock fluids are slower and dextrose-containing; dehydration is treated with ReSoMal; hypoglycaemia threshold is higher (<3.0 mmol/L); hypothermia is common.
- **Count every benzodiazepine dose**, including those given at home or at the referring clinic.

- **Common student mistakes**: queueing a child with emergency signs for registration; giving 20 mL/kg boluses to every febrile child; forgetting glucose; forgetting to weigh; using adult doses or adult vital signs; giving three or four doses of diazepam (respiratory arrest); not reassessing after each intervention.

EXAM FAVOURITE

Viva favourite: **"List the emergency signs."** Say them in ABCD order and include the details: obstructed/absent breathing, severe respiratory distress, central cyanosis, signs of shock (all three), coma, convulsing, and severe dehydration in a child with diarrhoea (two signs).

Red flags

RED FLAG

- **Bradycardia** or slow, gasping breathing in a sick child = imminent arrest: ventilate with bag and mask now.
- **Silent chest** or exhaustion in asthma or bronchiolitis.
- **Hypotension** in a child = late, decompensated shock.
- **Petechial/purpuric rash with fever** = meningococcal sepsis until proved otherwise.
- **Fluid overload during resuscitation**: rising RR, new crackles, enlarging liver, gallop rhythm, raised JVP in older children; **stop fluids**.
- Seizures not stopping after **two doses** of benzodiazepine (established status epilepticus).
- Coma with **unequal pupils, abnormal posturing or irregular breathing** = raised intracranial pressure.

Behind each emergency sign: the common causes

Emergency sign	Common causes	Distinguishing features	Key bedside test
Stridor / obstructed airway	Croup, foreign body, diphtheria (unimmunised), retropharyngeal abscess, anaphylaxis	Sudden choking episode → foreign body; barking cough → croup; grey membrane → diphtheria	Look in the mouth only if safe
Severe respiratory distress	Pneumonia, bronchiolitis, asthma, heart failure (anaemia, RHD, congenital), **acidosis** (DKA, severe malaria, dehydration)	Deep sighing breathing with clear chest = **metabolic acidosis**, not pneumonia	SpO_2, glucose, Hb, urine ketones
Shock	Septic shock, hypovolaemia (diarrhoea, burns, bleeding), anaphylaxis, cardiogenic, DKA	Diarrhoea/vomiting history; rash/urticaria; big heart and liver → cardiogenic	Glucose, Hb, mRDT
Coma	Cerebral malaria, meningitis, **hypoglycaemia**, post-ictal state, poisoning, head injury, DKA, encephalitis, hypertensive encephalopathy (AGN)	Recovery within 1 h → post-ictal; neck stiffness/bulging fontanelle → meningitis; smell of acetone → DKA	Glucose, mRDT, LP when safe
Convulsions	Febrile convulsion, cerebral malaria, meningitis, hypoglycaemia, epilepsy, electrolytes (hyponatraemia, hypocalcaemia), poisoning (OP), tetanus (spasms, conscious)	Simple febrile convulsion: 6 mo–5 y, generalised, <15 min, recovers fully	Glucose, mRDT, LP
Severe dehydration	Acute watery diarrhoea, cholera, dysentery, DKA (polyuria)	Polyuria + acidotic breathing + glucose high → DKA	Glucose, urine ketones
Severe pallor	Malaria, sickle cell crisis, hookworm/iron deficiency, malignancy, HIV	Splenomegaly, jaundice, haemoglobinuria (blackwater)	Hb, mRDT, sickling test

Scenarios

- **This febrile child has a CRT of 4 seconds but warm hands and a good pulse. Bolus?** No. That is impaired circulation, not shock. Give maintenance fluids (or feeds), treat the cause, and reassess every 15–30 minutes.
- **A wasted child with oedema is cold, with a weak fast pulse.** SAM shock protocol: 15 mL/kg over 1 hour of Ringer's lactate with 5% dextrose, pulse and RR every 5–10 minutes, glucose, warmth; never a 20 mL/kg saline bolus.
- **A convulsing child has had diazepam at the clinic and one dose here, and is still fitting.** That is two doses: go to **phenobarbital**, not a third benzodiazepine; bag-mask ready.
- **A child receiving antivenom or ceftriaxone develops stridor and urticaria.** Stop the infusion; **IM adrenaline** first; crystalloid if shocked.

6. Common Problems and Their Management

Airway and breathing

1. **Not breathing / gasping**: open airway, **5 rescue breaths** with bag and mask (correct mask size covering mouth and nose), check for signs of life; if none, start CPR (BLS below).
2. **Choking (foreign body)**:
   - Infant: **5 back blows** (infant face-down along your forearm, head lower than body), then **5 chest thrusts** (two fingers on the lower sternum); repeat; check the mouth and remove visible objects only.
   - Child >1 year: **5 back blows**, then **5 abdominal thrusts** (Heimlich) standing behind the child; repeat.
   - If the child becomes unresponsive, start CPR.
3. **Severe respiratory distress or cyanosis**: oxygen; position of comfort (let the child sit on the mother's lap); suction secretions gently; do not force an upset child with stridor to lie down.
4. **Specific treatment**: salbutamol for wheeze, nebulised adrenaline/dexamethasone for croup, antibiotics for pneumonia (all covered in their own topics).

Oxygen

Situation	Start oxygen if	Target
Child with emergency signs (obstructed breathing, severe distress, cyanosis, shock, coma, convulsions)	SpO_2 **<94%**	≥94% during resuscitation
Stable child without emergency signs	SpO_2 **<90%**	≥90%
No pulse oximeter	Central cyanosis, severe indrawing, grunting, RR ≥70, head nodding, unable to drink	Clinical improvement

Device	Flow
Nasal prongs	Neonate **0.5–1 L/min**; infant **1–2 L/min**; older child **2–4 L/min**
Face mask / head box	Higher flows (5–10 L/min); use in emergencies and with bag-mask ventilation
Bubble CPAP (where available)	Infants with severe respiratory distress not improving on prongs

Shock in a child without SAM (WHO ETAT 2016)

1. Oxygen; IV or IO access; take bloods (glucose, Hb, malaria test, cross-match, culture).
2. **Check for severe anaemia** (Hb <5 g/dL or severe pallor): if present, **give blood (20 mL/kg whole blood or 10 mL/kg packed cells)** rather than repeated crystalloid.
3. Otherwise give **Ringer's lactate or 0.9% saline 10–20 mL/kg over 30–60 minutes**.
4. **Reassess** after the bolus: pulse, CRT, hands, consciousness, RR, liver size, crackles.

5. Still in shock and **no** signs of fluid overload → give a second **10–20 mL/kg** over 30–60 min.
6. Still in shock after **40 mL/kg** → think septic shock needing inotropes (referral level), concealed bleeding, or cardiogenic shock; transfuse if Hb is falling; call senior.
7. Improved → **maintenance fluids** (or oral feeding) and treat the cause (antibiotics, antimalarial).
8. **Stop fluids at once** if the child develops fluid overload: crackles, rising RR, enlarging liver, gallop rhythm.

Shock in a child with SAM

Children with severe wasting or oedema have a weak heart and leaky cells; fast saline drowns them.

1. Oxygen, keep warm, check glucose (treat if <3.0 mmol/L).
2. IV **15 mL/kg over 1 hour** of **Ringer's lactate with 5% dextrose** (or half-strength Darrow's with 5% glucose, or 0.45% saline with 5% glucose).
3. Count pulse and RR **every 5–10 minutes**.
4. **Improving** (pulse and RR falling) → repeat **15 mL/kg over 1 hour**, then switch to **oral/NG ReSoMal 10 mL/kg/h** for up to 10 hours, then start F-75.
5. **Not improving after two boluses of 15 mL/kg** → assume **septic shock**: give **maintenance IV fluid (4 mL/kg/h)** while waiting for blood, then **transfuse fresh whole blood 10 mL/kg slowly over 3 hours**, start antibiotics, and refer/call senior.
6. **Stop** if RR rises by 5 or pulse by 15 per minute, or the liver enlarges (overload).

Severe anaemia with distress

- Transfuse if **Hb <4 g/dL**, or **Hb 4–6 g/dL with respiratory distress, shock, impaired consciousness or high parasitaemia**.
- **Whole blood 20 mL/kg** (or packed cells 10 mL/kg) over **3–4 hours**; in heart failure give 10 mL/kg packed cells and consider **furosemide 1 mg/kg IV**.
- In SAM: **10 mL/kg** over 3 hours only.

Coma and convulsions

1. **Position**: recovery position (if no neck injury); keep the airway clear; suction; oxygen.
2. **Glucose**: check; if **<2.5 mmol/L** (<3.0 in SAM), or if you cannot check, give **10% dextrose 5 mL/kg IV** (neonate 2 mL/kg) over 2–3 minutes, recheck in 30 minutes. If no IV: sugar or 10% dextrose via NG tube (Hypoglycaemia topic).
3. **Convulsing now**: give a benzodiazepine (table), time it; if still fitting after **10 minutes**, give a second dose; if still fitting after the second dose, give **phenobarbital** (or phenytoin) and call senior.
4. **Look for the cause**: fever, mRDT, neck stiffness, LP when stable, poisoning, trauma, hypertension (AGN).
5. **Treat fever** (paracetamol 15 mg/kg) and **raised ICP** (head up 30°, avoid fluid overload, avoid hypoxia).

LP timing: after the child is stable: not in shock, no signs of raised ICP (unequal pupils, abnormal posture, irregular breathing), not still convulsing, and no bleeding tendency. **Do not delay antibiotics for the LP.**

Anticonvulsant drug table

Drug	Dose	Route	Frequency	Max / notes
Diazepam	**0.5 mg/kg**	Rectal (draw up, remove the needle, insert the syringe 4–5 cm, hold buttocks together for a few minutes)	Once; repeat once after 10 min	Max **10 mg** per dose; max **2 doses** total; watch for apnoea
Diazepam	**0.2–0.3 mg/kg**	Slow IV over 1–2 min	Once; repeat once after 10 min	Max 10 mg; have a bag-mask ready

Drug	Dose	Route	Frequency	Max / notes
Midazolam	**0.3–0.5 mg/kg** (or age bands: 1–4 y 5 mg; 5–9 y 7.5 mg; ≥10 y 10 mg)	Buccal/intranasal	Once; repeat once after 10 min	Max **10 mg**; useful when no IV access
Lorazepam	**0.1 mg/kg**	IV	Once; repeat once after 10 min	Max 4 mg; rarely stocked in Uganda
Phenobarbital	Child **15–20 mg/kg** (WHO pocket book 15 mg/kg; UCG 2023 10–15 mg/kg in severe malaria); neonate **20 mg/kg**	IV over 15–20 min (or IM if no IV)	Loading dose once; neonates may get further 10 mg/kg (max 40 mg/kg total)	**First-line in neonates**; maintenance 5 mg/kg/day; watch for respiratory depression
Phenytoin	**15–20 mg/kg** (WHO pocket book 15–18 mg/kg)	IV over 20 min in 0.9% saline only, separate line	Loading once	Cardiac monitoring; not for neonates in most Ugandan units

DRUG DOSING

No more than two doses of a benzodiazepine (including any given at home or at the referring clinic). A third dose causes respiratory arrest, especially after phenobarbital. Neonates: use **phenobarbital 20 mg/kg** first, not diazepam.

Severe dehydration (no SAM): Plan C

Give **100 mL/kg Ringer's lactate** (or 0.9% saline if RL unavailable) IV:

Age	First give 30 mL/kg in	Then give 70 mL/kg in
Infant <12 months	**1 hour**	**5 hours**
Child ≥12 months	**30 minutes**	**2½ hours**

- Repeat the first 30 mL/kg once if the radial pulse is still very weak or absent.
- Reassess every 15–30 minutes; as soon as the child can drink, give **ORS 5 mL/kg/h** alongside.
- Reassess at 3 h (child) or 6 h (infant) and choose Plan A, B or C again.
- **Zinc** once rehydrated (doses in section 7).
- No IV access: ORS by NG tube **20 mL/kg/h for 6 hours** (120 mL/kg) and refer.
- **In SAM**: do not use Plan C. Use **ReSoMal 5 mL/kg every 30 min for 2 hours, then 5–10 mL/kg/h for up to 10 hours**, unless the child is in shock (SAM shock protocol).

Anaphylaxis

Recognise: sudden onset after exposure (drug, food, insect sting, antivenom, vaccine) with **airway** (stridor, swelling), **breathing** (wheeze, distress) or **circulation** (shock) problems, usually with skin signs (urticaria, flushing, angioedema).

1. Remove the trigger (stop the infusion or drug). Call for help.
2. **Adrenaline IM immediately** (table), anterolateral thigh. Repeat every **5 minutes** if not improving.
3. Lie flat with legs raised (sitting up if breathing is the problem; never stand the child up).
4. Oxygen; IV/IO access; **crystalloid 10 mL/kg** bolus if shocked, repeat as needed.
5. After two IM doses with continuing shock or airway compromise → senior/anaesthetist; IV adrenaline infusion only by experienced staff.
6. Second-line: **chlorphenamine** (for itch/urticaria) and **hydrocortisone** (may prevent late symptoms). These **never replace adrenaline**.

7. Observe at least **6–12 hours** (biphasic reactions); document the trigger and tell the family.

Drug	Dose	Route	Frequency	Max / notes
Adrenaline 1 mg/mL (1:1000)	**0.01 mg/kg = 0.01 mL/kg** (age bands: <6 months 0.1–0.15 mL; 6 months–6 years 0.15 mL; 6–12 years 0.3 mL; >12 years 0.5 mL)	IM, anterolateral thigh	Every 5 min if needed	Max **0.5 mg (0.5 mL)** per dose
0.9% saline / Ringer's lactate	**10 mL/kg**	IV/IO bolus	Repeat per response	Watch for overload
Chlorphenamine	6 months–6 years 2.5 mg; 6–12 years 5 mg; >12 years 10 mg	IV slowly or IM	Once, then oral	Second-line only
Hydrocortisone	**4 mg/kg** (6 months–6 years 50 mg; 6–12 years 100 mg; >12 years 200 mg)	IV/IM	Once	Second-line
Salbutamol	2.5 mg (<5 years) or 5 mg (≥5 years)	Nebulised	Repeat as needed	For persistent wheeze after adrenaline

Paediatric basic life support (BLS)

Based on the **2025 ILCOR consensus** as implemented by the **ERC 2025 and AHA 2025 paediatric life support guidelines** (the core sequence is unchanged from 2021: 5 rescue breaths, 15:2).

1. **Unresponsive?** Shout for help.
2. **Open the airway** (infant: neutral; child: head tilt–chin lift).
3. **Look, listen, feel ≤10 s**: breathing normally?
4. Not breathing or gasping → **5 rescue breaths** (bag–mask with oxygen if available).
5. **Signs of life / pulse >60?** (≤10 s). If not → **15 chest compressions : 2 breaths**, rate **100–120/min**, depth **one-third of the chest** (about 4 cm infant, 5 cm child).
6. **Alone?** After 1 minute of CPR go or phone for help.
7. Continue until signs of life, help arrives, or exhaustion.

Feature	Infant (<1 year)	Child (1 year–puberty)
Pulse check	Brachial	Carotid or femoral
Compression technique	Two thumbs encircling the chest (preferred); a lone rescuer may use two fingers (ERC) or the heel of one hand (AHA 2025 no longer recommends the two-finger technique)	Heel of one hand (or two hands in a bigger child), lower half of sternum
Depth	At least 1/3 of the chest, about **4 cm**	At least 1/3, about **5 cm**
Ratio	**15:2** (health workers); 30:2 acceptable for a lone lay rescuer	15:2
Rate	**100–120 per minute**	100–120 per minute

- **Pulse <60/min with poor perfusion in an infant = start compressions.**
- **Advanced life support** (hospital): adrenaline **10 µg/kg (0.1 mL/kg of 1:10,000) IV/IO every 3–5 minutes** for non-shockable rhythms; defibrillation **4 J/kg** for VF/pulseless VT. Reversible causes: the **4 Hs** (hypoxia, hypovolaemia, hypo/hyperkalaemia, hypothermia; also hypoglycaemia) and **4 Ts** (tension pneumothorax, tamponade, toxins, thromboembolism).
- **Intraosseous access** if no IV after 2 attempts or 90 seconds: proximal tibia, 1–2 cm below the tibial tuberosity on the medial flat surface; all fluids and drugs can go through it.

- Newborns follow **Helping Babies Breathe / neonatal resuscitation** (3:1 ratio, ventilation first); see the Neonatal Resuscitation topic.

Priority-sign children: what next

- **Tiny baby**: assess for sepsis, hypothermia, hypoglycaemia, jaundice; admit if sick.
- **Very high temperature**: paracetamol 15 mg/kg, mRDT, look for focus.
- **Poisoning**: see the Common Poisoning topic; do not induce vomiting.
- **Burns**: assess % body surface area; fluids if >10% (Parkland: 3–4 mL/kg × %BSA in 24 h plus maintenance), analgesia, tetanus.
- **Severe pallor**: Hb, malaria test, cross-match.
- **Malnutrition/oedema**: SAM protocol (glucose, warmth, antibiotics, F-75).
- **Referral**: read the note, re-triage, ask what was given (doses!).

Drug and fluid quick table (weight-based)

Drug / fluid	Dose	Route	Frequency	Max / notes
10% dextrose	**5 mL/kg** (neonate 2 mL/kg)	IV over 2–3 min	Recheck glucose at 30 min; repeat if low	Make 10% from 50%: 1 part 50% + 4 parts water for injection (e.g. 10 mL + 40 mL), or 10 mL of 50% in 90 mL of 5% dextrose
Ringer's lactate / 0.9% saline (shock, no SAM)	**10–20 mL/kg**	IV/IO over 30–60 min	Repeat once if still shocked and no overload	Total 40 mL/kg then senior review
RL + 5% dextrose (shock in SAM)	**15 mL/kg**	IV over 1 h	Repeat once if improving	Then ReSoMal
Whole blood	**20 mL/kg** (10 mL/kg in SAM/heart failure)	IV over 3–4 h	Once; recheck Hb	Packed cells 10 mL/kg
Ceftriaxone	**50 mg/kg** (80–100 mg/kg/day for meningitis)	IV/IM	Once daily (meningitis 100 mg/kg/day, once daily or divided 12-hourly)	Max 2 g/dose (4 g/day in meningitis)
Artesunate	**3 mg/kg** if <20 kg; **2.4 mg/kg** if ≥20 kg	IV/IM	0, 12, 24 h then daily	At least 3 doses, then oral ACT when able
Paracetamol	**15 mg/kg**	Oral/rectal	6-hourly	Max 60 mg/kg/day
Furosemide	**1 mg/kg**	IV	Once (heart failure/overload)	Max 40 mg

Complications of emergency care

Complication	Cause	Prevention
Fluid overload, pulmonary oedema	Fast boluses in malaria, sepsis, anaemia, SAM	Use the shock definition; 10–20 mL/kg over 30–60 min; reassess
Respiratory depression/arrest	Repeated benzodiazepines, phenobarbital	Max 2 benzodiazepine doses; bag-mask ready

Complication	Cause	Prevention
Aspiration pneumonia	Unconscious child lying supine, feeding by mouth	Recovery position; NG tube; nil by mouth
Hypoglycaemic brain injury	Missed glucose check	Check in every coma/convulsion; treat if unable to check
Transfusion reactions, overload	Fast or excess blood	20 mL/kg over 3–4 h; observe
Death in the queue	No triage	Triage before registration; re-triage the queue
Hypothermia	Exposure during resuscitation, SAM	Cover, skin-to-skin, warm room

7. Guidelines, Policy and Law in Uganda

- **Ministry of Health ETAT adoption**: ETAT is the national standard for paediatric emergency care, taught at Mulago, Mengo and RRHs. Triage happens **at the door, before registration or payment**.
- **Uganda Clinical Guidelines 2023, Chapter 1 (Emergencies and Trauma)**: the national reference for shock, anaphylaxis, convulsions, hypoglycaemia, dehydration and poisoning.
- **WHO ETAT guideline update (2016)**: WHO's current paediatric emergency guideline; cautious boluses after FEAST.
- **WHO Pocket Book of Hospital Care for Children (2013)**, Chapter 1 (triage and emergency conditions).
- **WHO wasting and nutritional oedema guideline (2023)**: SAM shock and dehydration protocols.
- **Resuscitation**: ILCOR 2025 consensus, ERC 2025 and AHA 2025 paediatric life support guidelines.
- **Level of care**: HC II/III stabilise (oxygen, glucose, first-dose antibiotics and antimalarial, diazepam, fluids) and **refer with a note and a health worker**; HC IV and General Hospitals admit, transfuse and run oxygen; RRHs and Mulago provide HDU/ICU, inotropes and blood gas.

UGANDA CONTEXT

Fluid boluses: UCG versus WHO. UCG 2023 and the WHO Pocket Book (2013) say **20 mL/kg Ringer's lactate or 0.9% saline "as rapidly as possible"** for shock (UCG's hypovolaemic-shock section 1.1.2.1, mainly bleeding and fluid loss, words it as 20–30 mL/kg over 60 minutes, repeated up to 60 mL/kg, then blood). The **WHO ETAT 2016** update, influenced by FEAST, recommends **10–20 mL/kg over 30–60 minutes** with reassessment, and **no bolus** for impaired circulation without full shock. Both say **15 mL/kg over 1 hour** in SAM. In practice: febrile shock (malaria/sepsis) → ETAT 2016 cautious boluses; haemorrhage or fluid loss → UCG volumes plus blood. Ask your senior which chart Mengo's emergency area uses; in an exam, quote ETAT 2016 and mention FEAST.

UGANDA CONTEXT

Zinc in diarrhoea: UCG versus WHO. The **WHO 2024 pneumonia and diarrhoea guideline** now suggests **5 mg daily for up to 14 days** (all ages up to 10 years); **UCG 2023** and most Ugandan wards still use **10 mg (<6 months) or 20 mg (≥6 months) daily for 10–14 days**. Know both and say which your ward follows.

UGANDA CONTEXT

Phenobarbital dose: WHO Pocket Book 15 mg/kg; UCG 2023 lists 10–15 mg/kg in severe malaria; many units use 15–20 mg/kg. Neonates 20 mg/kg first-line.

8. Key Points

High-yield points

- Triage **before registration or payment**: E, P or Q in 15–20 seconds.
- Treat in **ABCD order** before you know the diagnosis; **assess, treat, reassess**.
- Shock = **all three** of cold hands, CRT >3 s, weak fast pulse.
- **FEAST**: no boluses for impaired circulation; 10–20 mL/kg over 30–60 min only for true shock.
- SAM shock: **15 mL/kg over 1 hour**, dextrose-containing; then ReSoMal or blood.
- **Don't Ever Forget Glucose** in coma and convulsions.
- **Maximum two** benzodiazepine doses; then phenobarbital. Neonates: phenobarbital first.
- Severe anaemia with distress needs **blood**, not saline.
- Anaphylaxis: **IM adrenaline first**.
- Paediatric arrests are usually secondary: **5 rescue breaths**, then 15:2.

Numbers & doses

Item	Value
Triage time per child	**15–20 seconds**, before registration
Shock definition	Cold hands + **CRT >3 s** + weak fast pulse (all three)
Coma	AVPU **P or U**
Fast breathing	<2 mo **≥60**; 2–12 mo **≥50**; 1–5 y **≥40** /min
Oxygen threshold	SpO_2 **<94%** with emergency signs; **<90%** otherwise
Nasal prong flow	Neonate 0.5–1; infant 1–2; child 2–4 L/min
Shock bolus (no SAM)	**10–20 mL/kg** RL/NS over **30–60 min**; reassess
Shock in SAM	**15 mL/kg/h** RL + 5% dextrose; transfuse **10 mL/kg over 3 h** if no response
Hypoglycaemia	**<2.5 mmol/L** (SAM **<3.0**) → **10% dextrose 5 mL/kg** (neonate 2 mL/kg)
Diazepam	**0.5 mg/kg rectal** or **0.2–0.3 mg/kg IV**; max **10 mg**; max **2 doses**
Midazolam (buccal)	**0.3–0.5 mg/kg**, max 10 mg
Phenobarbital	**15–20 mg/kg IV/IM** over 15–20 min (neonate 20 mg/kg)
Plan C	100 mL/kg: infant 30 in 1 h + 70 in 5 h; child 30 in ½ h + 70 in 2½ h
Transfusion	Hb **<4**, or **<6 with distress**: whole blood **20 mL/kg** over 3–4 h
Anaphylaxis	Adrenaline 1:1000 **0.01 mL/kg IM**, max 0.5 mL, every 5 min
BLS	5 breaths, then **15:2**, 100–120/min, depth 1/3 of the chest
ALS adrenaline	**10 µg/kg** (0.1 mL/kg of 1:10,000) IV/IO every 3–5 min; defibrillation **4 J/kg**
Weight estimate	1–5 y: **(2 × age) + 8 kg**

Exam traps

- **"Register and pay, then see the doctor."** Wrong: triage happens **before** registration; emergency cases are treated first.
- **"Febrile child with cold hands: 20 mL/kg saline stat."** Wrong unless **all three** shock signs are present; FEAST showed harm.

- **"SAM child in shock: 20 mL/kg normal saline bolus."** Wrong: **15 mL/kg over 1 hour** of RL with 5% dextrose.
- **"Third dose of diazepam for continuing seizures."** Wrong: after **two doses**, go to **phenobarbital**.
- **"Diazepam first for a convulsing neonate."** Wrong: **phenobarbital 20 mg/kg**.
- **"Hydrocortisone and chlorphenamine for anaphylaxis."** Wrong as first line: **IM adrenaline** first.
- **"Severe anaemia with heart failure: give saline for the tachycardia."** Wrong: **slow blood transfusion** ± furosemide.
- **"Unconscious child: send the malaria test and wait."** Wrong: **check and treat glucose** now.

Questions seniors ask

Viva questions

1. *What are the ETAT triage categories?* Emergency (ABCD signs, treat immediately), Priority (assess next: 3 TPR-MOB), Queue.
2. *Define shock in ETAT.* Cold hands **plus** capillary refill >3 s **plus** weak fast pulse. All three.
3. *Why not give everyone a 20 mL/kg bolus?* FEAST (Uganda, Kenya, Tanzania) showed boluses increased mortality in febrile children with impaired perfusion; give boluses only for true shock, 10–20 mL/kg over 30–60 min, reassessing.
4. *How do you manage shock in SAM?* 15 mL/kg Ringer's lactate with 5% dextrose over 1 hour; repeat if improving then ReSoMal; if not improving, maintenance fluid and transfuse 10 mL/kg over 3 hours.
5. *A child is convulsing and you have no IV access. What do you give?* Rectal diazepam 0.5 mg/kg (max 10 mg) or buccal midazolam; plus glucose (sugar/10% dextrose via NG if no IV); repeat the benzodiazepine once after 10 minutes.
6. *When do you give oxygen?* SpO_2 <94% if emergency signs, <90% if stable; or clinical signs (central cyanosis, severe indrawing, grunting, head nodding) if no oximeter.
7. *First drug in anaphylaxis, and dose?* IM adrenaline 1:1000, 0.01 mL/kg (max 0.5 mL), anterolateral thigh, repeat every 5 minutes.
8. *What is the BLS compression:ventilation ratio for children?* 15:2 for health workers, after 5 rescue breaths, at 100–120/min.

On the ward round

- **What are the emergency signs?** Obstructed/absent breathing, severe respiratory distress, central cyanosis, signs of shock (all three), coma or convulsing, and severe dehydration (2 signs) in a child with diarrhoea.
- **This febrile child has a CRT of 4 seconds but warm hands and a good pulse. Bolus?** No. That is impaired circulation, not shock. Give maintenance fluids (or feeds), treat the cause, and reassess every 15–30 minutes.
- **How would you make 10% dextrose if you only have 50%?** Mix 1 part 50% dextrose with 4 parts water for injection (e.g. 10 mL + 40 mL), or 10 mL of 50% in 90 mL of 5% dextrose.
- **When would you do an LP in this convulsing child?** After the child is stable: not in shock, no signs of raised ICP (unequal pupils, abnormal posture, irregular breathing), not still convulsing, and no bleeding tendency. Do not delay antibiotics for the LP.
- **What do you do if you cannot get an IV line in a shocked child?** Intraosseous needle in the proximal tibia (1–2 cm below the tibial tuberosity, medial flat surface) after 2 failed attempts or 90 seconds; all fluids and drugs can go through it.
- **Why are children's arrests usually secondary?** Because they come from respiratory failure or shock, which is why BLS starts with rescue breaths and why early ABC treatment prevents arrest.
- **How do you estimate the weight of a 3-year-old with no scale?** (2 × 3) + 8 = 14 kg (APLS); weigh as soon as possible, and use a length-based tape (Broselow) if available.

Mnemonics

- **ABCD**: Airway-Breathing, Circulation, Coma-Convulsions, Dehydration.
- **3 TPR-MOB**: Tiny baby, Temperature, Trauma; Pallor, Poisoning, Pain; Respiratory distress, Restless/irritable/lethargic, Referral; Malnutrition, Oedema, Burns.
- **Shock = "Cold, Slow, Weak"**: cold hands, slow capillary refill (>3 s), weak fast pulse.
- **"DEFG: Don't Ever Forget Glucose"** in coma and convulsions.
- **Plan C "30–70"**: 30 mL/kg then 70 mL/kg; infants 1 h + 5 h, children ½ h + 2½ h.
- **SAM shock "15-15-ReSoMal"**: 15 mL/kg/h, repeat 15 if improving, then ReSoMal.
- **Anaphylaxis "0.01 in the thigh"**: 0.01 mL/kg of 1:1000 IM.
- **BLS "5-15-2"**: 5 rescue breaths, then 15 compressions to 2 breaths.

OSCE tips

- In an ETAT station, **speak aloud**: "I am checking for emergency signs: airway and breathing, circulation, coma and convulsions, dehydration."
- **Weigh or estimate weight first** and show the calculation for every dose.
- **Reassess** after every intervention and say what you are looking for.
- Ask for **glucose** in every unconscious or convulsing child.
- Don't forget to **keep the child warm** and to **talk to the mother**.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	What do the letters of ETAT stand for, and what are its three triage categories?	**Emergency Triage, Assessment and Treatment**. Categories: **Emergency** (treat now), **Priority** (assess next, ahead of the queue), **Queue** (non-urgent).
2	List the ETAT emergency signs (ABCD).	**A/B**: obstructed or absent breathing, severe respiratory distress, central cyanosis. **C**: signs of shock. **C**: coma or convulsing. **D**: severe dehydration in a child with diarrhoea.
3	WHO definition of shock in a child?	**All three**: cold hands **+** capillary refill **>3 seconds +** weak, fast pulse.
4	Priority signs mnemonic?	**3 TPR-MOB**: Tiny baby (<2 months), Temperature very high, Trauma; Pallor (severe), Poisoning, Pain (severe); Respiratory distress, Restless/irritable/lethargic, Referral (urgent); Malnutrition (visible severe wasting), Oedema of both feet, Burns (major).
5	What is AVPU and which levels count as coma?	**A**lert, responds to **V**oice, responds to **P**ain, **U**nresponsive. **P or U = coma** (about GCS ≤8).
6	Signs of severe dehydration in a child with diarrhoea (ETAT)?	Any **two** of: lethargy or unconsciousness, sunken eyes, very slow skin pinch (≥2 s), unable to drink or drinks poorly.
7	Oxygen thresholds in ETAT 2016?	Start oxygen if SpO_2 **<94%** in a child with emergency signs; **<90%** once stable without emergency signs.
8	Nasal prong oxygen flow rates by age?	Neonate **0.5–1 L/min**; infant **1–2 L/min**; older child **2–4 L/min**.
9	Fluid bolus for shock in a child WITHOUT severe malnutrition (WHO ETAT 2016)?	**10–20 mL/kg** Ringer's lactate or 0.9% saline over **30–60 minutes**, reassess, repeat once if still shocked and no overload. Transfuse if severely anaemic.
10	Fluid for shock in a child WITH severe acute malnutrition?	**15 mL/kg over 1 hour** of Ringer's lactate with 5% dextrose (or half-strength Darrow's with 5% glucose, or 0.45% saline with 5% glucose). Pulse and RR every 5–10 min. Improving → repeat 15 mL/kg, then ReSoMal 10 mL/kg/h. No improvement after two boluses → maintenance 4 mL/kg/h and fresh whole blood **10 mL/kg over 3 h**.
11	What did the FEAST trial show and where was it done?	In febrile African children with impaired perfusion (Uganda, Kenya, Tanzania), **fluid boluses increased 48-hour mortality** (about 10.6% vs 7.3%) compared with no bolus. It is why ETAT 2016 restricts boluses to true shock and gives them more slowly.
12	Hypoglycaemia threshold and treatment in ETAT?	Glucose **<2.5 mmol/L** (45 mg/dL), or **<3.0 mmol/L** (54 mg/dL) in SAM, or unable to measure in a child with coma/convulsions → **10% dextrose 5 mL/kg IV** (neonate 2 mL/kg); recheck after 30 min.
13	How do you make 10% dextrose from 50% dextrose?	1 part 50% + **4 parts water for injection** (e.g. 10 mL + 40 mL), or 10 mL of 50% in **90 mL of 5% dextrose**. Never give undiluted 50% to a child.
14	Diazepam doses for a convulsing child?	**Rectal 0.5 mg/kg** or **slow IV 0.2–0.3 mg/kg**; max **10 mg** per dose; repeat once after 10 min; **no more than two doses** in total.
15	Second-line anticonvulsant after two benzodiazepine doses?	**Phenobarbital 15–20 mg/kg IV/IM over 15–20 min** (WHO pocket book 15 mg/kg in children; 20 mg/kg in neonates), or phenytoin 15–20 mg/kg IV over 20 min. Call senior.
16	First-line anticonvulsant in a neonate?	**Phenobarbital 20 mg/kg IV** (further 10 mg/kg doses up to 40 mg/kg total), not diazepam. Check glucose first.

#	Question	Answer
17	Plan C fluid volumes and timing?	Ringer's lactate **100 mL/kg**: infant <12 months **30 mL/kg in 1 h, then 70 mL/kg in 5 h**; child ≥12 months **30 mL/kg in 30 min, then 70 mL/kg in 2½ h**.
18	Transfusion thresholds and volume in ETAT?	Hb **<4 g/dL**, or **4–6 g/dL with distress, shock or impaired consciousness** → whole blood **20 mL/kg** (packed cells 10 mL/kg) over **3–4 h**; in SAM 10 mL/kg over 3 h.
19	Adrenaline dose for anaphylaxis in a child?	Adrenaline **1 mg/mL (1:1000) 0.01 mL/kg IM** into the anterolateral thigh, max 0.5 mL; repeat every 5 min. Age bands (UCG 2023): <6 years 0.15 mL, 6–12 years 0.3 mL, >12 years 0.5 mL.
20	Paediatric BLS sequence for health workers?	Open airway → check breathing ≤10 s → **5 rescue breaths** → check signs of life → **15 compressions : 2 breaths** at **100–120/min**, depth **one-third of the chest** (about 4 cm infant, 5 cm child).
21	Choking infant: what do you do?	Infant face-down along your forearm, head low: **5 back blows**, then turn and give **5 chest thrusts**; repeat. Remove only visible objects. If unresponsive, start CPR. (Abdominal thrusts are for children over 1 year.)
22	APLS weight estimate for a child aged 1–5 years?	**(2 × age in years) + 8 kg**. Infants: (0.5 × age in months) + 4. Age 6–12: (3 × age) + 7.
23	Fast breathing cut-offs (WHO)?	<2 months **≥60**/min; 2–12 months **≥50**/min; 1–5 years **≥40**/min.
24	Signs of fluid overload during resuscitation?	RR rises (by ≥5/min), pulse rises (by ≥15/min), new fine crackles, enlarging liver, gallop rhythm, raised JVP. **Stop the infusion.**
25	When should you NOT do an LP in a convulsing, febrile child yet?	Signs of raised ICP (unequal pupils, abnormal posture, irregular breathing, deep coma), shock, still convulsing, or bleeding tendency. Give antibiotics first; LP when stable.
26	Where do you put an intraosseous needle in a small child, and when?	**Proximal tibia**, flat medial surface 1–2 cm below the tibial tuberosity, after **2 failed IV attempts or 90 seconds** in an emergency. Any drug or fluid can go through it.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. At Mengo outpatients, the triage nurse sees five children who arrived together. Which child must be taken into the resuscitation area FIRST?

A A 4-year-old with a fever of 39.8 °C who is alert and playing
B A 3-week-old baby who is breastfeeding well but has a mild umbilical discharge
C A 2-year-old with diarrhoea who is lethargic, has sunken eyes and a very slow skin pinch
D A 6-year-old with severe abdominal pain who is walking
E A 10-month-old with visible severe wasting who is alert

2. A 3-year-old girl (estimated 14 kg) with 2 days of fever is brought to a HC IV. Her hands are cold, capillary refill is 4 seconds and the radial pulse is weak at 168/min. She is AVPU V. Hb 10 g/dL, no diarrhoea, no wasting or oedema. What is the MOST appropriate fluid management?

A Ringer's lactate 10–20 mL/kg (140–280 mL) over 30–60 minutes, then reassess
B 0.9% saline 40 mL/kg as fast as possible
C No fluid bolus; maintenance fluids only because FEAST showed boluses are harmful
D Ringer's lactate with 5% dextrose 15 mL/kg over 1 hour
E Whole blood 20 mL/kg over 3 hours

3. A 2-year-old boy (11 kg) with fever is brought to Mengo by his mother. He is warm-handed, alert, capillary refill 3.5 seconds, pulse 150/min with good volume. mRDT positive, Hb 8.9 g/dL. A student suggests 20 mL/kg saline. What is the BEST response? Advanced

A Agree; any prolonged capillary refill needs a bolus
B Give 10 mL/kg saline over 15 minutes as a compromise
C Give 15 mL/kg of Ringer's lactate with 5% dextrose over 1 hour
D No bolus; give oral fluids or maintenance fluids, treat malaria, and reassess
E Transfuse 20 mL/kg whole blood

4. A 14-month-old (7.2 kg, MUAC 10.9 cm, bilateral pedal oedema +) is lethargic with cold hands, capillary refill 4 s and a weak pulse of 170/min. Glucose 3.6 mmol/L. Which is the MOST appropriate initial fluid?

A 0.9% saline 20 mL/kg (144 mL) as rapidly as possible
B Ringer's lactate with 5% dextrose 15 mL/kg (about 110 mL) over 1 hour, checking pulse and RR every 5–10 minutes
C ReSoMal 5 mL/kg orally every 30 minutes
D Plan C: 30 mL/kg Ringer's lactate over 1 hour then 70 mL/kg over 5 hours
E No IV fluid; start F-75 feeds every 2 hours

5. A 4-year-old (16 kg) is convulsing on arrival. The mother says a clinic gave 'an injection in the bottom' (diazepam) 20 minutes ago and he has fitted continuously since. You have just given rectal diazepam 8 mg 10 minutes ago, and glucose is 4.8 mmol/L. He is still convulsing. What is the NEXT step?

A Give a third dose of diazepam 0.5 mg/kg rectally
B Give 10% dextrose 5 mL/kg IV
C Wait another 10 minutes because diazepam takes time to work
D Give paracetamol and tepid sponge, then reassess
E Give phenobarbital 15–20 mg/kg IV over 15–20 minutes (or IM) and call a senior

6. A 3-year-old (13 kg) with 3 days of fever is unrousable (AVPU U) on arrival at a HC IV. He is not convulsing, SpO_2 95%, warm hands. The glucometer has no strips. What should you do FIRST after securing the airway and positioning him?

A Give 10% dextrose 65 mL IV
B Give rectal diazepam 6.5 mg
C Do a lumbar puncture
D Send blood for a malaria smear and wait for the result
E Give IV ceftriaxone and refer immediately without other treatment

7. Which statement about oxygen in ETAT (WHO 2016) is CORRECT?

A Give oxygen to every child with fever and fast breathing
B Target SpO_2 88–92% in all children
C Children with emergency signs should receive oxygen if SpO_2 is below 94%
D Nasal prongs for a neonate should run at 4 L/min
E Without a pulse oximeter, oxygen should never be given

8. A 9-month-old (8 kg) has had profuse watery diarrhoea for 2 days. She is lethargic, eyes sunken, skin pinch goes back very slowly; no wasting or oedema; radial pulse present. IV access is obtained. What is the correct Plan C regimen?

A Ringer's lactate 240 mL over 30 minutes, then 560 mL over 2½ hours
B ORS 75 mL/kg over 4 hours
C Ringer's lactate with 5% dextrose 120 mL over 1 hour
D 0.9% saline 160 mL over 15 minutes and repeat as needed
E Ringer's lactate 240 mL over 1 hour, then 560 mL over 5 hours

9. A 7-year-old (22 kg) develops generalised urticaria, lip swelling, wheeze and a weak pulse 5 minutes after IV ceftriaxone. What is the FIRST drug to give?

A Hydrocortisone 100 mg IV
B Adrenaline 1:1000, 0.22 mL IM into the anterolateral thigh
C Chlorphenamine 5 mg IV
D Adrenaline 1:1000, 2.2 mL IV push
E Nebulised salbutamol 5 mg

10. A 2-year-old (12 kg) with cerebral malaria has Hb 3.8 g/dL, RR 60, an enlarged liver and fine basal crackles. Pulse 170, hands warm, capillary refill 2 s. What is the MOST appropriate management of the circulation?

Advanced

A 0.9% saline 20 mL/kg bolus over 15 minutes
B Transfuse 20 mL/kg whole blood over 30 minutes
C No transfusion until the malaria is treated
D Transfuse packed cells 10 mL/kg (or whole blood) slowly over 3–4 hours, consider furosemide 1 mg/kg IV, and monitor for overload
E Ringer's lactate with 5% dextrose 15 mL/kg over 1 hour

11. A 10-month-old is found choking on a groundnut. He is coughing weakly, cannot cry and is becoming blue, but is still conscious. What should you do?

- A Give 5 back blows with the infant face-down along your forearm, then 5 chest thrusts, and repeat
- B Perform 5 abdominal thrusts (Heimlich manoeuvre)
- C Do a blind finger sweep of the mouth
- D Encourage him to keep coughing and wait
- E Start 15:2 CPR immediately

12. You find a 5-year-old (18 kg) unresponsive and not breathing on the ward. You have shouted for help and opened the airway. According to paediatric BLS, what comes next?

- A Start 30 chest compressions immediately
- B Check for a pulse for 30 seconds
- C Give 5 rescue breaths, then check for signs of life
- D Leave the child to fetch the emergency trolley
- E Give adrenaline 0.1 mL/kg of 1:10,000 IV

13. A 6-month-old (6 kg) is brought with 5 minutes of continuous jerking of all limbs and fever. IV access fails twice. Glucose 5.0 mmol/L. Which drug and dose is MOST appropriate now?

- A Phenobarbital 20 mg/kg orally
- B Diazepam 0.3 mg/kg intramuscularly
- C Diazepam 10 mg rectally
- D Diazepam 3 mg (0.5 mg/kg) rectally
- E Phenytoin 20 mg/kg rectally

14. During a 20 mL/kg bolus for a 4-year-old with septic shock, the RR rises from 40 to 52, the liver edge moves from 2 cm to 5 cm below the costal margin and crackles appear. What is the BEST action? Advanced

- A Speed up the bolus because the child is still tachycardic
- B Stop the infusion, sit the child up, give oxygen, consider furosemide and call a senior
- C Switch to Ringer's lactate with 5% dextrose at the same rate
- D Finish the bolus and then reassess
- E Give a second 20 mL/kg bolus of colloid

15. At a HC III, a 2-year-old (11 kg) with severe malaria is convulsing; IV access is impossible and there is no glucometer. Which combination is BEST before referral? Advanced

- A Oral paracetamol and oral artemether-lumefantrine
- B Intramuscular quinine and oral glucose by mouth
- C IV fluids by scalp vein and wait for the ambulance
- D Referral immediately without any treatment to save time
- E Rectal diazepam 0.5 mg/kg, sugar or 10% dextrose via nasogastric tube (or sublingual sugar), and rectal artesunate (or IM artesunate) before referral

Answers & explanations

> **1. C** · A 2-year-old with diarrhoea who is lethargic, has sunken eyes and a very slow skin pinch
>
> C is correct: lethargy plus sunken eyes plus a very slow skin pinch in a child with diarrhoea is **severe dehydration**, an emergency (D) sign needing Plan C now. A is wrong: very high temperature is a priority sign, not an emergency sign. B is wrong: a tiny baby (<2 months) is a priority sign; this baby is feeding well. D is wrong: severe pain is a priority sign. E is wrong: visible severe wasting is a priority sign (M of 3 TPR-MOB); without emergency signs the child is assessed next, not resuscitated.

2. A · Ringer's lactate 10–20 mL/kg (140–280 mL) over 30–60 minutes, then reassess

A is correct: she has **all three** signs of shock and is not malnourished or severely anaemic, so WHO ETAT 2016 advises **10–20 mL/kg isotonic crystalloid over 30–60 min** with reassessment. B is wrong: 40 mL/kg rapid boluses are the regimen FEAST showed increased mortality. C is wrong: FEAST applies to impaired circulation without full shock; a child in true shock still needs a bolus. D is wrong: 15 mL/kg of RL-dextrose over 1 hour is the SAM shock protocol. E is wrong: Hb 10 g/dL does not need transfusion.

3. D · No bolus; give oral fluids or maintenance fluids, treat malaria, and reassess

D is correct: he has only one sign (prolonged CRT), so he has **impaired circulation, not shock**. After FEAST, he should get **no bolus**, but maintenance or oral fluids, antimalarial treatment and repeated assessment. A is wrong: WHO requires all three signs for shock. B is wrong: a smaller bolus is still a bolus that FEAST showed harmful in this group. C is wrong: that is the SAM shock regimen. E is wrong: Hb 8.9 g/dL without distress does not need blood.

4. B · Ringer's lactate with 5% dextrose 15 mL/kg (about 110 mL) over 1 hour, checking pulse and RR every 5–10 minutes

B is correct: this child has **SAM (MUAC <11.5 cm, oedema) and shock**; WHO/UCG give **15 mL/kg over 1 hour** of RL with 5% dextrose (or half-strength Darrow's/0.45% saline with glucose), monitoring for overload. A is wrong: rapid saline boluses cause heart failure in SAM. C is wrong: ReSoMal is for dehydration without shock. D is wrong: Plan C is contraindicated in SAM. E is wrong: a shocked child needs IV fluid now; feeding comes after stabilisation.

5. E · Give phenobarbital 15–20 mg/kg IV over 15–20 minutes (or IM) and call a senior

E is correct: he has already had **two benzodiazepine doses** (one at the clinic, one here) and is in established status epilepticus; the next drug is **phenobarbital** (WHO pocket book 15 mg/kg; or phenytoin), with airway support ready. A is wrong: a third benzodiazepine dose risks respiratory arrest. B is wrong: glucose is normal. C is wrong: continuing seizures beyond 5–10 minutes after a benzodiazepine need second-line treatment now. D is wrong: antipyretics do not stop a seizure.

6. A · Give 10% dextrose 65 mL IV

A is correct: in a child with coma, **if glucose cannot be measured, treat as hypoglycaemia**: 10% dextrose **5 mL/kg** (13 × 5 = 65 mL). A is the life-saving, low-risk step. B is wrong: he is not convulsing. C is wrong: an LP in deep coma risks herniation and does not come first. D is wrong: never wait for results to treat a reversible cause; IV artesunate also follows. E is wrong: ceftriaxone may well be needed, but referral without glucose (and artesunate) misses the most urgent treatment.

7. C · Children with emergency signs should receive oxygen if SpO_2 is below 94%

C is correct: during resuscitation, children with emergency signs get oxygen to keep SpO_2 **≥94%**; stable children get oxygen if SpO_2 <90%. A is wrong: fast breathing alone does not mean hypoxaemia. B is wrong: 88–92% is a target used in some adult COPD patients, not children. D is wrong: neonates need 0.5–1 L/min by prongs. E is wrong: without an oximeter, oxygen is given for clinical signs such as central cyanosis, grunting, severe indrawing or head nodding.

8. E · Ringer's lactate 240 mL over 1 hour, then 560 mL over 5 hours

E is correct: an infant under 12 months gets **30 mL/kg in 1 hour** (240 mL) then **70 mL/kg in 5 hours** (560 mL). A is wrong: those timings are for children ≥12 months. B is wrong: ORS 75 mL/kg over 4 h is Plan B (some dehydration). C is wrong: that is the SAM shock protocol. D is wrong: rapid repeated saline boluses are not Plan C and ignore the structured 100 mL/kg plan.

9. B · Adrenaline 1:1000, 0.22 mL IM into the anterolateral thigh

B is correct: anaphylaxis is treated with **IM adrenaline 0.01 mL/kg of 1:1000** (22 × 0.01 = 0.22 mL; UCG band 6–12 years = 0.3 mL), repeated every 5 minutes, after stopping the ceftriaxone. A is wrong: hydrocortisone is slow and second-line. C is wrong: antihistamines treat itch and urticaria only. D is wrong: 2.2 mg IV adrenaline is a 10-fold, potentially fatal overdose and 1:1000 is never pushed IV. E is wrong: salbutamol helps residual wheeze but does not treat shock or airway swelling.

10. D · Transfuse packed cells 10 mL/kg (or whole blood) slowly over 3–4 hours, consider furosemide 1 mg/kg IV, and monitor for overload

D is correct: this is **severe anaemia with heart failure** (Hb <4, tachypnoea, big liver, crackles); the treatment is **blood given slowly**, packed cells 10 mL/kg (or whole blood) over 3–4 h, with furosemide if overloaded. A is wrong: crystalloid worsens heart failure and dilutes Hb further. B is wrong: blood over 30 minutes is for haemorrhagic shock, not compensated anaemia. C is wrong: Hb <4 g/dL needs transfusion now, alongside IV artesunate. E is wrong: that regimen is for shock in SAM.

11. A · Give 5 back blows with the infant face-down along your forearm, then 5 chest thrusts, and repeat

A is correct: an infant with an **ineffective cough** needs **5 back blows then 5 chest thrusts**, repeated until the object comes out or he becomes unresponsive. B is wrong: abdominal thrusts are for children over 1 year (risk of liver injury in infants). C is wrong: blind finger sweeps can push the object deeper; remove only visible objects. D is wrong: waiting is only for an effective (loud, forceful) cough. E is wrong: CPR starts if he becomes unresponsive.

12. C · Give 5 rescue breaths, then check for signs of life

C is correct: paediatric arrests are usually **hypoxic**, so BLS gives **5 rescue breaths** first, then checks signs of life (≤10 s) and starts **15:2** compressions if absent. A is wrong: 30:2 starting with compressions is adult-style BLS. B is wrong: pulse checks must take no more than 10 seconds. D is wrong: a lone health worker does 1 minute of CPR before leaving to get help; here help has been called. E is wrong: adrenaline comes in advanced life support after CPR has started and access is obtained.

13. D · Diazepam 3 mg (0.5 mg/kg) rectally

D is correct: without IV access, give **rectal diazepam 0.5 mg/kg** (6 × 0.5 = 3 mg), repeat once after 10 min if still fitting; buccal midazolam is an alternative. A is wrong: oral drugs cannot be given to a convulsing child. B is wrong: IM diazepam is erratically absorbed and slow. C is wrong: 10 mg is the maximum dose, an overdose for a 6 kg infant. E is wrong: phenytoin is not given rectally.

14. B · Stop the infusion, sit the child up, give oxygen, consider furosemide and call a senior

B is correct: rising RR, an enlarging liver and new crackles are **fluid overload**; stop the fluid at once, support breathing, consider furosemide 1 mg/kg and get senior help (inotropes may be needed). A is wrong: more fluid will cause pulmonary oedema. C is wrong: changing the fluid type does not fix overload. D is wrong: you stop the bolus as soon as overload appears. E is wrong: colloid adds volume and offers no advantage.

15. E · Rectal diazepam 0.5 mg/kg, sugar or 10% dextrose via nasogastric tube (or sublingual sugar), and rectal artesunate (or IM artesunate) before referral

E is correct: stop the seizure (**rectal diazepam**), treat presumed hypoglycaemia without IV access (**NG sugar solution/10% dextrose 5 mL/kg** or a moistened teaspoon of sugar under the tongue), and give **pre-referral artesunate** (rectal or IM), then refer with a health worker. A is wrong: nothing can be swallowed safely during a convulsion, and oral ACT is not for severe malaria. B is wrong: oral glucose risks aspiration, and quinine is second-line and causes hypoglycaemia. C is wrong: IV access has failed; waiting delays life-saving steps. D is wrong: pre-referral treatment saves lives.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The convulsing toddler in the queue

You are the junior clerk in the Mengo paediatric emergency area at 09:30. A mother rushes in carrying an 18-month-old boy who 'started shaking' in the registration queue 5 minutes ago. He has had fever for 3 days. He is jerking all four limbs, drooling, and his lips look dusky.

1. How do you triage him and what are your first actions?

> **Emergency**: convulsing (C) and possible cyanosis (B). Carry him to the resuscitation bed, call for help, note the time. Airway: clear secretions with gentle suction, place him on his side (recovery position), **oxygen** by face mask. Do not put anything in his mouth. Estimate weight with the APLS 1–5 year formula: (2 × 1.5) + 8 = **11 kg** (weigh as soon as he is stable).

2. IV access is not yet available. What drug do you give and how much?

> **Rectal diazepam 0.5 mg/kg = 5.5 mg** (round to 5 mg) using a 1 mL syringe with the needle removed, inserted 4–5 cm, buttocks held together. Note the time; if still fitting after 10 min, repeat once (or give IV 0.2–0.3 mg/kg once access is in). Ask the mother whether he had any anticonvulsant before arriving.

3. What must you check at the same time, and how will you act on it?

> **Blood glucose**. If <2.5 mmol/L (or cannot be measured): **10% dextrose 5 mL/kg = 55 mL IV** once access is in; if no access, sugar solution/10% dextrose via NG tube. Also take blood for **mRDT/smear, Hb, cross-match**. Here glucose is **1.9 mmol/L**, mRDT positive, Hb 6.1 g/dL.

4. The seizure stops after the first diazepam dose and the dextrose. He is AVPU P, SpO_2 95% on oxygen, warm hands, CRT 2 s. What is your working diagnosis and immediate treatment?

> **Severe malaria** (cerebral malaria/impaired consciousness, hypoglycaemia, moderate-severe anaemia); differential **bacterial meningitis**. Give **IV artesunate 3 mg/kg = 33 mg** (<20 kg) at 0, 12, 24 h then daily; recheck glucose in 30 min and give **maintenance fluids with dextrose** (no bolus: not shocked); consider **ceftriaxone 100 mg/kg/day** until meningitis is excluded; paracetamol 15 mg/kg for fever; recovery position, NG tube, nil by mouth while P.

5. What will you monitor over the next 24 hours, and when would you do an LP?

> Blantyre/AVPU and seizures hourly, **glucose 3–4 hourly** (and with any deterioration), RR, SpO_2, pulse, temperature, urine output, Hb (transfuse if <4 g/dL or <6 with distress), fluid balance. **LP** once he is stable with no signs of raised ICP (if coma persists beyond the post-ictal period or neck stiffness appears). Check HIV status and immunisation.

6. Present him on the ward round in two sentences.

> "An 18-month-old boy, 11 kg, triaged Emergency for a generalised convulsion with fever for 3 days; he had hypoglycaemia (1.9 mmol/L), a positive mRDT and Hb 6.1 g/dL; the seizure stopped after one dose of rectal diazepam and 10% dextrose. Working diagnosis: severe malaria with impaired consciousness and hypoglycaemia; on IV artesunate, maintenance dextrose fluids and ceftriaxone pending LP."

Take-away: Treat the convulsion and the glucose before the diagnosis; no more than two benzodiazepine doses; in Uganda, every febrile convulsing child needs a malaria test and consideration of meningitis.

Case 2: Shock in a malnourished child

At a regional referral hospital, a 20-month-old girl is brought with 4 days of diarrhoea and vomiting. She weighs 6.8 kg, MUAC 10.6 cm, no oedema. She is lethargic, hands cold, capillary refill 5 s, radial pulse weak at 172/min, RR 48.

1. How do you triage her and what are the key problems?

> **Emergency**: signs of **shock** (all three), lethargy, likely severe dehydration, in a child with **SAM** (MUAC <11.5 cm). SAM changes the fluid plan, the glucose threshold (<3.0 mmol/L) and the risk of hypothermia and sepsis.

2. What are your immediate steps in the first 5 minutes?

> Oxygen, IV access (or IO), **glucose** (treat if <3.0 mmol/L with 10% dextrose 5 mL/kg = 34 mL), blood for Hb, malaria test, cross-match and culture; keep her **warm** (cover, skin-to-skin, warm room); call senior.

3. Prescribe her IV fluid.

> **15 mL/kg over 1 hour = about 100 mL** of **Ringer's lactate with 5% dextrose** (or half-strength Darrow's with 5% glucose, or 0.45% saline with 5% glucose). Count **pulse and RR every 5–10 minutes**. **Not** Plan C and **not** 20 mL/kg saline.

4. After 1 hour her pulse is 150, RR 44, she is more alert. What next?

> She is improving with no overload: **repeat 15 mL/kg over 1 hour**, then switch to **ReSoMal 10 mL/kg/h orally/NG for up to 10 hours** (or low-osmolarity ORS if ReSoMal is unavailable, per WHO 2023), then start **F-75**. Start IV antibiotics (ampicillin + gentamicin) for presumed sepsis.

5. What if she had NOT improved after two boluses?

> Assume **septic shock**: give **maintenance IV fluid 4 mL/kg/h** while blood is arranged, then **fresh whole blood 10 mL/kg slowly over 3 hours**, IV antibiotics, and start F-75 when possible. **Stop** fluids if RR rises by 5 or pulse by 15, or the liver enlarges.

6. What are the next 'stabilisation' priorities in SAM?

> The WHO 10 steps: prevent/treat **hypoglycaemia** (feeds every 2–3 h day and night), **hypothermia**, **dehydration** (ReSoMal), **electrolytes**, **infection** (antibiotics), **micronutrients** (no iron in stabilisation), start cautious feeding (F-75), then catch-up growth, stimulation, and follow-up. Check HIV status.

Take-away: In SAM, shock fluids are slower (15 mL/kg/h), contain dextrose, and stop at the first sign of overload; if no response after two boluses, think sepsis and give blood.

Case 3: Collapse after an injection

A 9-year-old boy (25 kg) admitted with pneumonia receives IV ceftriaxone on the ward. Three minutes later he complains of itching, then becomes hoarse and breathless. You find him with facial swelling, widespread wheals, wheeze, pulse 150/min and cold hands.

1. What is the diagnosis and the first two actions?

> **Anaphylaxis** (skin signs plus airway, breathing and circulation involvement after a drug). (1) **Stop the ceftriaxone** and call for help. (2) **Adrenaline 1:1000 IM, 0.01 mL/kg = 0.25 mL** (UCG band 6–12 years: 0.3 mL) into the anterolateral thigh.

2. What position and supportive treatment do you give?

> Lie flat with legs raised if shocked (sitting if breathing is the main problem, but never standing), **high-flow oxygen**, IV access and **0.9% saline 10 mL/kg (250 mL)** bolus; nebulised salbutamol 5 mg for persisting wheeze.

3. After 5 minutes he is still wheezy and hypotensive. What now?

> **Repeat IM adrenaline** (doses every 5 minutes as needed), repeat fluid bolus, call anaesthesia/senior; if there is no response after two IM doses, an IV adrenaline infusion by experienced staff in a monitored area.

4. Which second-line drugs might you give, and why are they second?

> **Chlorphenamine** (6–12 years 5 mg IV/IM) for itch and urticaria, and **hydrocortisone** (6–12 years 100 mg IV) to reduce the risk of protracted symptoms. They act too slowly to treat airway swelling or shock, so they never replace adrenaline.

5. How do you manage him after recovery?

> Observe for at least **6–12 hours** (biphasic reaction), document **ceftriaxone allergy** clearly on the file and treatment chart, report the adverse drug reaction (NDA pharmacovigilance form), choose a non-beta-lactam antibiotic for his pneumonia after discussion, and explain to the family.

Take-away: Anaphylaxis = IM adrenaline 0.01 mL/kg of 1:1000 into the thigh, repeated every 5 minutes; antihistamines and steroids are extras, never the first drug.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.